

SEQUOIA CAPITAL ANALYSIS — SEPTEMBER 2023

Generative AI's Act Two: From Technology Hammer to Real Value

How the generative AI market is shifting from novelty demos to end-to-end solutions that solve real human problems

By Sonya Huang, Pat Grady and GPT-4



THE FOUNDATION

Six Decades of Technological Progress Converged to Enable Generative AI

60

Years of Moore's Law

Six decades of Moore's Law delivered the compute horsepower to process exaflops of data.

40

Years of the Internet

Four decades of the internet (accelerated by COVID) produced trillions of tokens of training data.



COMMERCIAL TRACTION

GenAI Startups Surpassed \$1B Revenue Faster Than SaaS Ever Did

> \$1B

Startup revenue in months, not years

11

Midjourney employees at hundreds of millions in revenue

2h

Average session time on Character.ai

ChatGPT

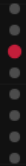
Fastest-growing application in history, with strong product-market fit among students and developers.

Midjourney

Hundreds of millions in revenue with a team of just 11 people.

Character.ai

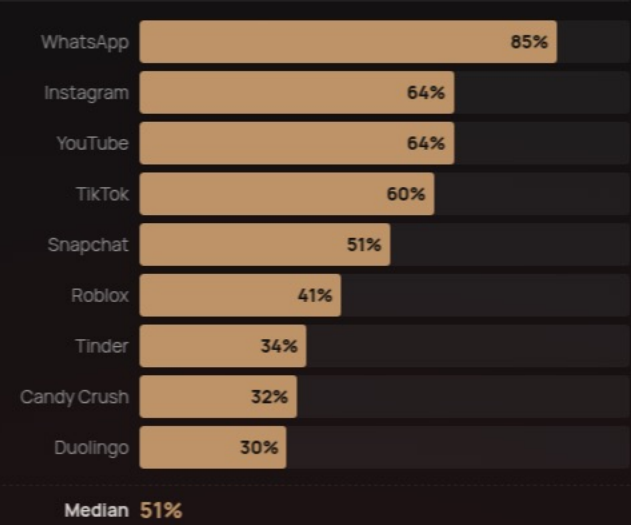
Popularized AI entertainment and companionship; users spend 2 hours per session in-app.



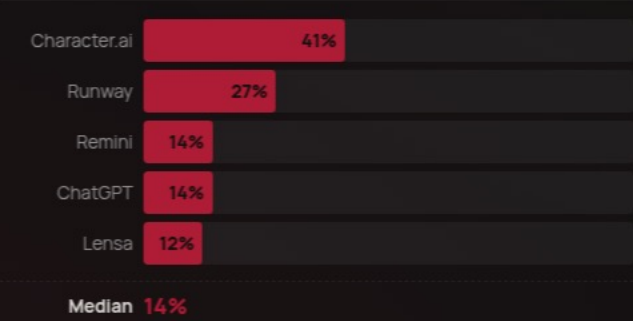
THE RETENTION PROBLEM

AI-First Apps Show 14% Median Retention vs. 51% for Incumbents

Incumbents

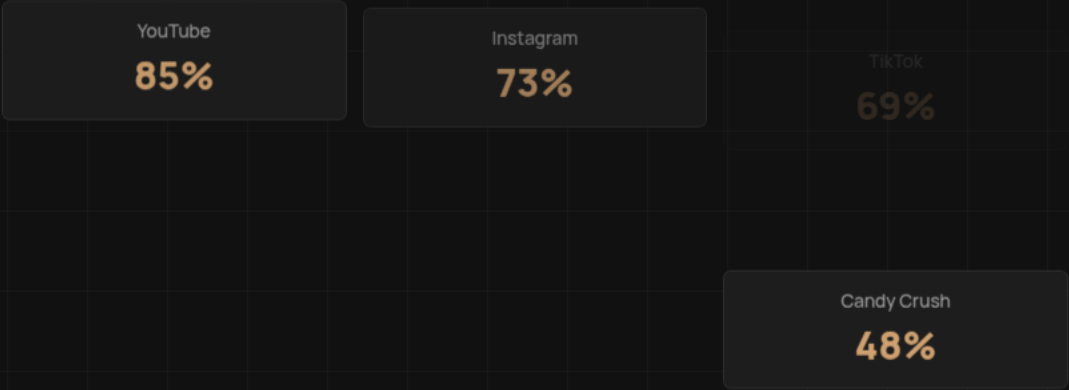


AI-First Companies



ENGAGEMENT BENCHMARK

YouTube Leads Incumbents at 85% Monthly Active Usage



MARKET TRANSITION

Act 1 Was Technology-Out; Act 2 Is Customer-Back

Act 1: Technology-Out

- Foundation models as the **entire solution**
- Novelty demos and lightweight wrappers
- Cool technology seeking a problem
- Poor user retention and stickiness
- Thin layer on top of existing APIs

Act 2: Customer-Back

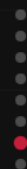
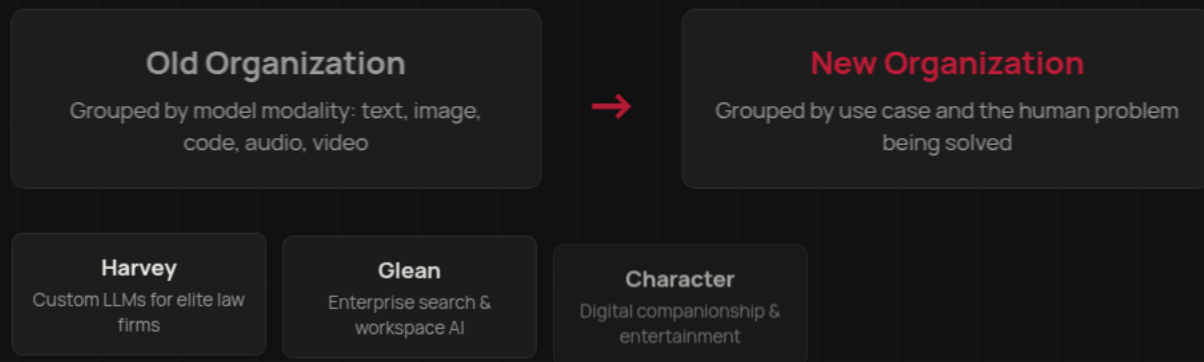
- Foundation models as **a piece** of a comprehensive solution
- New editing interfaces for stickier workflows
- **Multi-modal** by design
- End-to-end problem solving, not thin wrappers
- Human problems drive product architecture

Exemplars: Harvey, Glean, Character, Ava



STRUCTURAL SHIFT

Market Map Reorganized by Use Case, Not Model Modality



CONCLUSION

Key Takeaways: What Separates Winners in Act Two

- 1 Retention is the critical challenge** — AI-first apps at 14% median retention vs. 51% for incumbents. Companionship/entertainment (Character.ai at 41%) may be the strongest consumer use case.
- 2 Foundation models should be a component, not the whole product.** Act 2 winners integrate models into comprehensive, workflow-integrated solutions.
- 3 Multi-modal, end-to-end solutions will win.** New editing interfaces and multi-modal design create stickier workflows and better outputs.
- 4 The \$1B+ revenue milestone proves real demand exists** despite hype cycles. Generative AI had a more successful commercial start than SaaS.

